



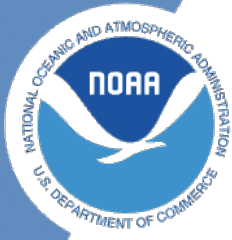
NOAA Technical Memorandum NMFS-XXX-##

NOAA quarto book

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National Marine Fisheries Service
Northwest Fisheries Science Center



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1 Introduction

Here we will be displaying various data summaries and analyses related to dolphin management in the U.S. South Atlantic.

1.1 Load Libraries

The documentation of this data exploration was developed in R, using R Studio and Quarto. The following R libraries were also used. The **pals** package allows for improved visualization with different color palettes (**pals?**). The **rerddap** package allows for efficient downloads of physical data from the ERDDAP server (**rerddap?**).

2 NOAA Fisheries landings data

Here we will explore commercial and recreational landings data accessed through the FOSS (Fisheries One Stop Shop) data portal. The advantage of this portal is that commercial and recreational data are available side-by-side in the same units (pounds landed).

2.1 Data access and upload

Data are downloaded from the NOAA Fisheries FOSS site. For the Data Set, we choose both commercial and recreational. For the Years, we select all years from 1980 to 2021. For NMFS Region, we select the "NMFS Regions" button and then select the Gulf, South Atlantic, Middle Atlantic and New England from the box. For the Species, we select "Dolphinfish" and "Dolphinfish**." For Report Format, we choose TOTALS BY YEAR/REGION. Hitting the "RUN REPORT" button then generates the data needed for the subsequent analysis, which can be downloaded under the "Actions" button.

We then read in the data file and clean it for analysis (removing the commas from the numbers so that R recognizes the values as numbers). We output the file header to make sure it was read in and that the numbers were converted correctly.

```
# clear workspace
rm(list = ls())

# read in data file
d <- read.table("data/FOSS_landings.csv", header = T, sep = ",", skip = 1)
head(d)
```

	Year	Region.Name	Pounds	Dollars	Collection	Metric.Tons
1	1980	Gulf	94,984	55,676	Commercial	43
2	1980	Middle Atlantic	400	256	Commercial	0
3	1980	New England	200	54	Commercial	0

2 NOAA Fisheries landings data

4	1980	South Atlantic	76,042	45,852	Commercial	34
5	1981	Gulf	73,075	46,951	Commercial	33
6	1981	Gulf	1,559,544		Recreational	707

```
# take commas out of file -----
d$Pounds <- as.numeric(gsub(",", "", d$Pounds))
d$Dollars <- as.numeric(gsub(",", "", d$Dollars))
d$Metric.Tons <- as.numeric(gsub(",", "", d$Metric.Tons))
head(d)
```

	Year	Region.Name	Pounds	Dollars	Collection	Metric.Tons
1	1980	Gulf	94984	55676	Commercial	43
2	1980	Middle Atlantic	400	256	Commercial	0
3	1980	New England	200	54	Commercial	0
4	1980	South Atlantic	76042	45852	Commercial	34
5	1981	Gulf	73075	46951	Commercial	33
6	1981	Gulf	1559544	NA	Recreational	707

2.2 Summarize the landings

We then summarize the landings (in pounds) by year, sector, and region. The result is a new data frame with each row representing a year and each column representing the total landings for each region and sector.

```
# compile by year for sectors and regions -----
d$Collection <- factor(d$Collection, levels = c("Recreational", "Commercial"))
tab <- tapply(d$Pounds, list(d$Year, d$Collection, d$Region.Name), sum, na.rm = T)
tab[is.na(tab)] <- 0
#head(tab)

dat <- data.frame(cbind(tab[, , "Gulf"], tab[, , "South Atlantic"],
                        tab[, , "Middle Atlantic"], tab[, , "New England"]))
names(dat) <- c("rec_Gulf", "com_Gulf", "rec_SA", "com_SA", "rec_MA", "com_MA", "rec_NE", "com_NE")
head(dat)
```

	rec_Gulf	com_Gulf	rec_SA	com_SA	rec_MA	com_MA	rec_NE	com_NE
1980	0	94984	0	76042	0	400	0	200
1981	1559544	73075	8421898	59037	0	0	0	0

2 NOAA Fisheries landings data

1982	4197503	162457	7573035	143960	2097	200	0	0
1983	2162936	190808	9134022	127986	45927	1400	0	400
1984	1189573	282976	7265430	158901	0	1700	0	400
1985	2542089	269477	6684821	142684	113666	5000	0	4800

2.3 Plot the summary

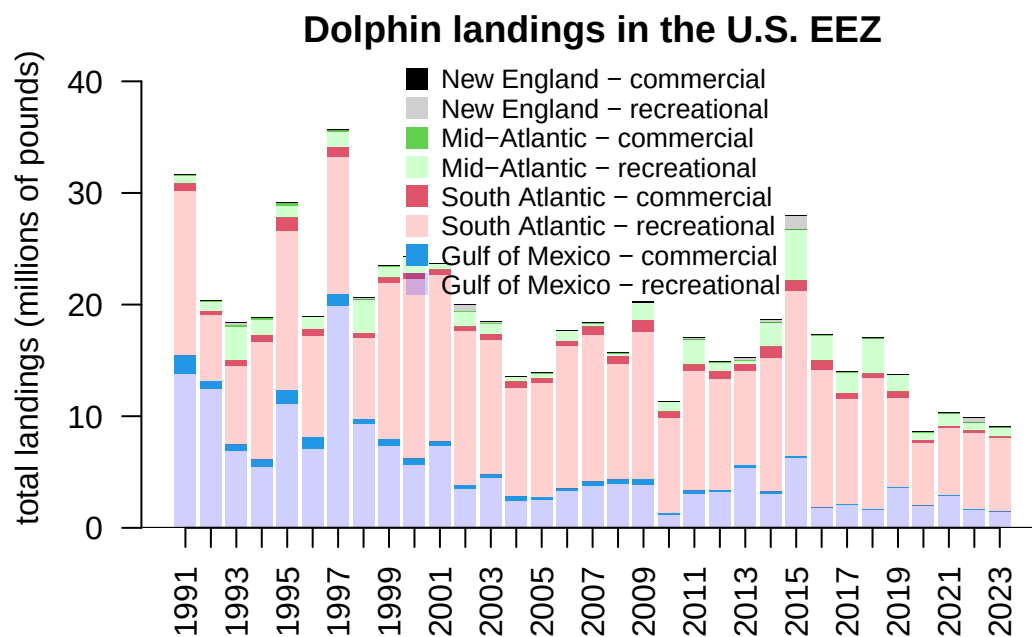
Now we plot the data, using similar color schemes by region. Because the recreational data have high imprecision prior to the 1990s we plot from 1990 to the last year of data.

```
# select years -----
dat <- dat[which(rownames(dat) > 1990), ]

# define colors and plot -----
cols <- c("#0000FF30", 4, "#FF000030", 2, "#00FF0030", 3, "#00000030", 1)

par(mar = c(4, 4, 1, 0), mgp = c(2.5, 1, 0))
yrs <- as.numeric(rownames(dat))

b <- barplot(t(dat)/10^6, names.arg = yrs, col = cols, ylim = c(0, 43),
             border = NA, las = 2, ylab = "total landings (millions of pounds)",
             main = "Dolphin landings in the U.S. EEZ")
legend("top", c("New England - commercial", "New England - recreational",
               "Mid-Atlantic - commercial", "Mid-Atlantic - recreational",
               "South Atlantic - commercial", "South Atlantic - recreational",
               "Gulf of Mexico - commercial", "Gulf of Mexico - recreational"),
       col = cols[8:1], pch = 15, pt.cex = 1.5, bty = "n", y.intersp = 0.9, cex = 0.85)
axis(1, at = b, lab = rep("", length(b)), pos = 0)
abline(h=0)
```



3 Updated Gulf/Atlantic parsing of data

There are some issues with the way that the FOSS system splits out Gulf and Atlantic catches by Monroe County, which is on the border between the two jurisdictions. We received data from Mike Larkin (SERO) which has the correct split out of data for Monroe County between the Gulf and Atlantic. We compare the total recreational catch for the Gulf and Atlantic between the FOSS data and the updated data. The numbers are very close; the average difference from the FOSS data is 2%.

```
ml <- read.table("data/RecreationalDolphinLandings_SAandGOM_MLarkin.csv", header = T, sep = ",")
ml <- ml[which(ml$Year > 1990), ]

# select same years for FOSS data set
dat <- dat[which(yrs <= max(ml$Year)), ]

# check that years match
summary(rownames(dat) == ml$Year)
```

```
      Mode      TRUE
logical      31
```

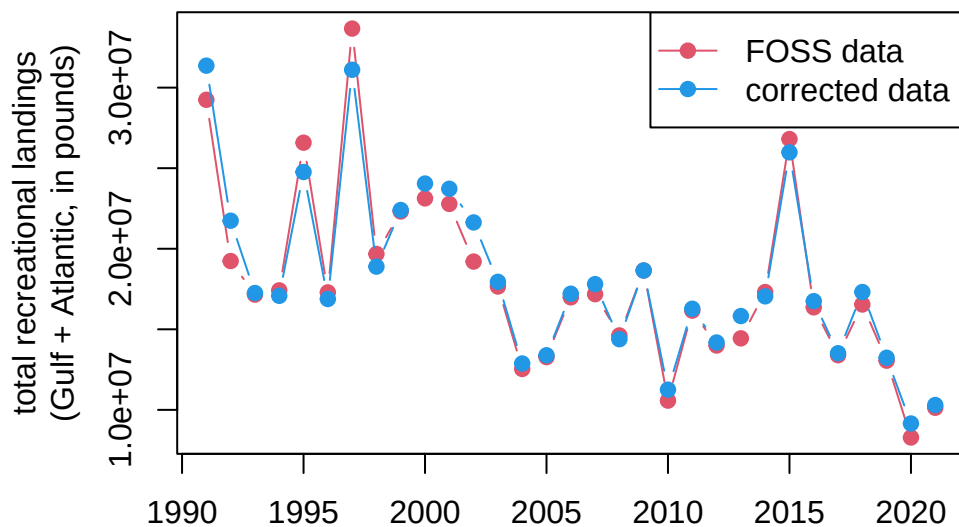
```
yrs <- as.numeric(rownames(dat))

# check that totals match
old <- dat$rec_Gulf + dat$rec_SA + dat$rec_MA + dat$rec_NE
new <- ml$GOM + ml$ATL

# plot totals
par(mar = c(4, 6, 2, 1), mgp = c(2.5, 1, 0))
matplot(yrs, cbind(old, new), xlab = "", lty = 1, col = c(2, 4), type = "b", pch = 19,
        ylab = "total recreational landings\n(Gulf + Atlantic, in pounds)",
        main = "Comparison of FOSS data with corrected split by Monroe County")
legend("topright", c("FOSS data", "corrected data"), col = c(2, 4), pch = 19, lty = 1)
```

3 Updated Gulf/Atlantic parsing of data

Comparison of FOSS data with corrected split by Monroe



```
# average % difference between FOSS and updated data  
mean((old - new)/old*100)
```

```
[1] -2.023886
```

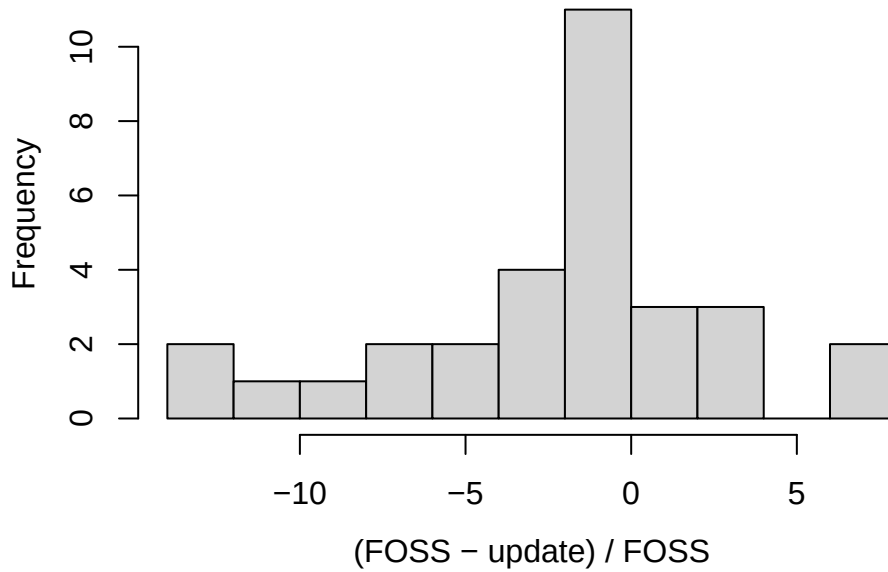
```
# min and max % difference between FOSS and updated data  
range((old - new)/old*100)
```

```
[1] -13.008552  7.587789
```

```
hist((old - new)/old*100, main = "Percent difference from FOSS data", breaks = 10,  
     xlab = "(FOSS - update) / FOSS")
```

3 Updated Gulf/Atlantic parsing of data

Percent difference from FOSS data



```
# replace FOSS data with update data
datnew <- dat
datnew$rec_Gulf <- ml$GOM
datnew$rec_SA <- ml$ATL - (datnew$rec_MA + datnew$rec_NE)

# old data set
tail(dat)
```

	rec_Gulf	com_Gulf	rec_SA	com_SA	rec_MA	com_MA	rec_NE	com_NE
2016	1777348	37411	12357817	828133	2212513	61336	16499	15747
2017	2064106	65141	9457460	532973	1820541	16179	48607	30969
2018	1626095	56188	11713866	435059	3104018	20514	97107	12667
2019	3567170	83956	7967978	563132	1518629	48224	66	16918
2020	1952241	26332	5653294	260311	679556	14040	0	1880
2021	2893817	26570	6061024	178822	1051785	25318	123038	11421

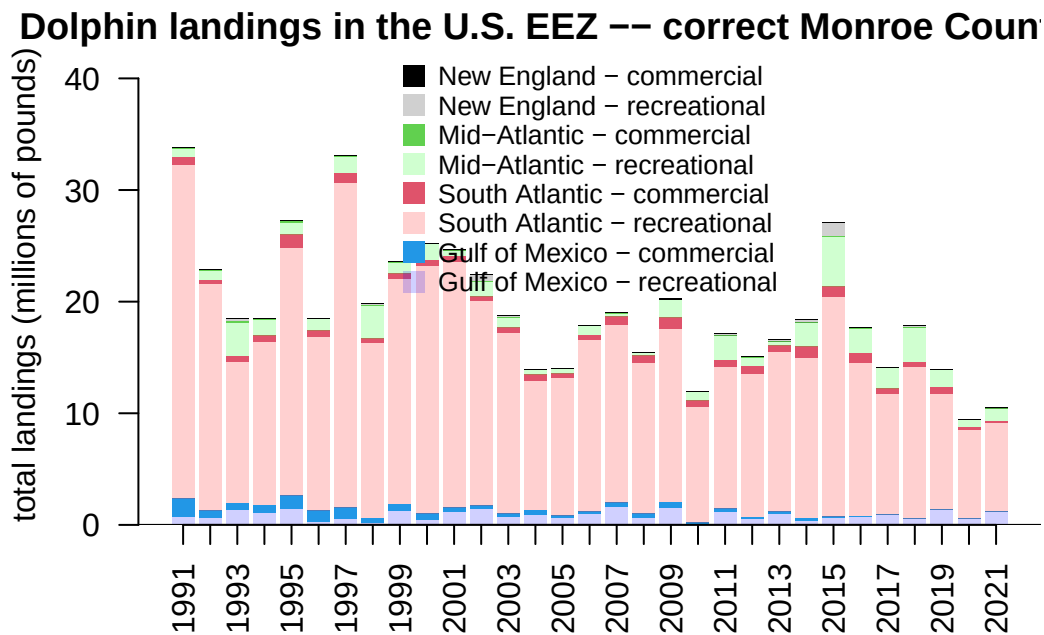
```
# updated data set
tail(datnew)
```

	rec_Gulf	com_Gulf	rec_SA	com_SA	rec_MA	com_MA	rec_NE	com_NE
--	----------	----------	--------	--------	--------	--------	--------	--------

3 Updated Gulf/Atlantic parsing of data

2016	751499	37411	13766851	828133	2212513	61336	16499	15747
2017	868489	65141	10777104	532973	1820541	16179	48607	30969
2018	508570	56188	13603875	435059	3104018	20514	97107	12667
2019	1304865	83956	10410599	563132	1518629	48224	66	16918
2020	555695	26332	7916891	260311	679556	14040	0	1880
2021	1174412	26570	7950716	178822	1051785	25318	123038	11421

We replace the FOSS values for Gulf and South Atlantic recreational catches with these new values and replot the full data set. We can also plot the landings as a line graph so that the relative magnitudes of catch across region and sector are more easily compared.



3 Updated Gulf/Atlantic parsing of data

